

A CERTIFIED FINITE VERIFICATION OF ERDŐS–SZEKERES #699 FOR $i = 3$ AND $n \leq 10^8$

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ABSTRACT. Erdős and Szekeres asked whether, for every $1 \leq i < j \leq n/2$, the two binomial coefficients $\binom{n}{i}$ and $\binom{n}{j}$ have a common prime divisor $p \geq i$. We give a certified finite result, not a proof of the full conjecture: the assertion holds for the complete stratum $i = 3$ and every $n \leq 100,000,000$. The certificate is an exact exhaustive enumeration based on the complete odd-prime support of $\binom{n}{3}$ and Lucas digit conditions. A second implementation reconstructs the same range using a different sieve and anchor choice and evaluates decisive divisibility by Legendre valuations. A direct arbitrary-precision gcd oracle checks all 994,009 admissible pairs through $n = 2000$. No proof assistant is used. Cases with $n > 10^8$ and the general strata $i \geq 4$ remain open.

1. INTRODUCTION

Erdős and Szekeres formulated the following question in 1978 [3]: for all integers

$$1 \leq i < j \leq \lfloor n/2 \rfloor,$$

does there exist a prime $p \geq i$ dividing both $\binom{n}{i}$ and $\binom{n}{j}$? Their stronger proposed inequality $p > i$ has exceptional triples; for example

$$\gcd\left(\binom{28}{5}, \binom{28}{14}\right) = 2^3 3^3 5,$$

so equality $p = i$ is essential.

The current Erdős Problems record lists the weak question as open and records partial results for $j \leq 3i/2$ and for $n = 2j$ [1]. One classical input to that line of work is the theorem of Ecklund, Eggleton, Erdős, and Selfridge comparing the parts of a binomial coefficient supported below and at least at i [2]. A public exact-arithmetic scanner reports no weak counterexample for any i through $n = 10^7$, and also treats two selected families at larger individual values of n [4].

This note certifies a different, sharply delimited endpoint.

Theorem 1.1. *For all integers n, j satisfying*

$$8 \leq n \leq 100,000,000, \quad 4 \leq j \leq \lfloor n/2 \rfloor,$$

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there is an odd prime p such that

$$p \mid \binom{n}{3} \quad \text{and} \quad p \mid \binom{n}{j}.$$

Consequently the weak Erdős–Szekeres assertion holds on the complete stratum $i = 3$ through $n = 10^8$.

The bound is a finite computational theorem. In the sources and database queries recorded with the code, we did not locate a previous all-row $i = 3$ verification at this endpoint. This is a bounded search statement, not a claim of absolute priority.

2. EXACT REDUCTION

For $n \geq 8$, define

$$A_n = \binom{n}{3} = \frac{n(n-1)(n-2)}{6}$$

and let S_n be the set of its odd prime divisors.

Lemma 2.1 (Complete prime support). *The set S_n consists of every prime $p > 3$ dividing at least one of $n, n-1, n-2$, together with 3 exactly when*

$$n \bmod 9 \in \{0, 1, 2\}.$$

Proof. No prime $p > 3$ is cancelled by the denominator 6. Among three consecutive integers exactly one is divisible by 3; a factor 3 remains after division by 3! exactly when that integer is divisible by 9. This is equivalent to the displayed residue condition. $\square$

We use the following digit criterion, including a proof to make the certificate endpoint self-contained.

Lemma 2.2 (Lucas digit criterion). *Let p be prime and write $n = \sum_r n_r p^r$ and $j = \sum_r j_r p^r$ in base p . Then*

$$p \nmid \binom{n}{j} \iff j_r \leq n_r \text{ for every } r.$$

Proof. In $\mathbb{F}_p[x]$, the freshman's dream gives $(1+x)^p = 1+x^p$. Iterating over the base- p digits of n and comparing the coefficient of x^j yields

$$\binom{n}{j} \equiv \prod_r \binom{n_r}{j_r} \pmod{p}.$$

Each factor is nonzero precisely when $j_r \leq n_r$. $\square$

For $a \in S_n$, let

$$T_a(n) = \{j : 4 \leq j \leq \lfloor n/2 \rfloor, j_r \leq n_r \text{ in base } a\}.$$

Proposition 2.3 (Finite decision at one row). *Fix $n \geq 8$ and any anchor $a \in S_n$. The assertion of Theorem 1.1 at this n holds if and only if every $j \in T_a(n)$ has some $p \in S_n$ with $p \mid \binom{n}{j}$.*

Proof. By Lemma 2.2, every $j \notin T_a(n)$ already has the common odd prime a . Hence only $T_a(n)$ can contain a counterexample. By Lemma 2.1, testing all $p \in S_n$ is exactly testing every eligible odd prime divisor of A_n ; no other prime can witness the desired gcd. $\square$

Proposition 2.3 is the completeness argument behind both programs. It is not a heuristic filter.

3. CERTIFICATE AND INDEPENDENT VERIFICATION

The discovery program performs the following exact finite procedure for every n in the certified interval.

- (1) Reconstruct S_n from Lemma 2.1 using a linear smallest-prime-factor sieve and a rolling three-integer window.
- (2) Choose an anchor minimizing the number of digit subchoices, enumerate $T_a(n)$ recursively, and test all remaining primes with the Lucas predicate of Lemma 2.2.
- (3) Stop and serialize a literal triple if a counterexample occurs; otherwise serialize a null result for the exact inclusive range.

The independent verifier does not import a coverage bitmap or any cached discovery result. It parses only the exact range from a strict seven-field certificate, rebuilds a classical Eratosthenes smallest-prime-factor table, refactors all three numerator terms separately at every row, and chooses the largest member of S_n as anchor. For every other prime it evaluates

$$v_p \binom{n}{j} = \sum_{r \geq 1} \left(\left\lfloor \frac{n}{p^r} \right\rfloor - \left\lfloor \frac{j}{p^r} \right\rfloor - \left\lfloor \frac{n-j}{p^r} \right\rfloor \right),$$

rather than calling the discovery program's Lucas evaluator.

Both complete runs returned a null counterexample and independently visited 43,631,335,536 anchor-compatible indices. The count and elapsed times are diagnostics, not proof assumptions. The decisive artifacts are listed in Table 1.

A third oracle directly constructs arbitrary-precision binomial coefficients and gcds for all 994,009 admissible pairs with $8 \leq n \leq 2000$; it also returns no counterexample. The strict verifier rejects incomplete JSON, duplicate-key JSON, and a certificate that reports a non-null witness before starting enumeration.

The C++ verifier reconstructs the range but does not consume the scanner's diagnostic count. Without repeating the enumeration, a fail-closed release binder parses the canonical JSON and strict output, pins their hashes, the source hash and build command, and requires the same endpoint $n \in [8, 10^8]$, stratum $i = 3$, and count 43,631,335,536. Runtime is ignored.

TABLE 1. SHA-256 hashes of decisive sources and outputs.

Artifact	SHA-256
scan_i3.cpp	4d082ab479f1afe267dd4eea26cbbbed03 de1038a3f053f108ec34b183b159338
verify_i3.cpp	5365f052cd14ae35efa5e7efed5008584 fb4ab523f8f1a6c51058a14a2d9bcd5
i3_scan_100m.json	23c45255e41877ccde381d86564e991ce d9b7e08b8559f48405f4a759b51f0ba
i3_verify_100m_strict.txt	9a1a430068f5813195a90b0f22937095 008b50f84c66fb311ca64ff3dc2904ab

External working-directory tests passed under normal Python and `python -O`; altered endpoints, hashes, counts, truncated output, and duplicate fields all failed closed.

Proof of Theorem 1.1. For each integer n in the stated range, the two independent exhaustive runs implement Proposition 2.3, using the complete support of Lemma 2.1. The strict verifier returned success for the entire inclusive range and no row produced a counterexample. Therefore every admissible j has a common odd prime divisor, which is a prime $p \geq 3$. $\square$

4. ADDITIONAL STRUCTURAL REDUCTIONS

The computation does not prove an infinite statement, but two elementary reductions explain the remaining search geometry. Let $V_i(n)$ be the part of $\binom{n}{i}$ supported on primes at least i . From

$$\binom{n}{i} \binom{n-i}{j-i} = \binom{n}{j} \binom{j}{i}$$

one obtains, in any counterexample,

$$(15) \quad V_i(n) \mid \binom{j}{i}.$$

Kummer's carry count gives $v_q(\binom{n}{i}) \leq \lfloor \log_q n \rfloor$ for every prime $q < i$. Consequently

$$V_i(n) \geq \frac{\binom{n}{i}}{n^{\pi(i-1)}}.$$

Combining this with (15) and $\binom{n}{i}/\binom{j}{i} \geq (n/j)^i$ yields the necessary condition

$$(16) \quad n^{i-\pi(i-1)} \leq j^i.$$

In particular, (16) proves the conjecture for $i = 1, 2$ and restricts a possible $i = 3$ counterexample to $n \leq j^{3/2}$.

There is also a prime-gap obstruction. If a prime p lies in $(n-i, n]$, then $p > n/2 \geq j$ and $n \bmod p < i < j$; the one-digit case of Lemma 2.2 makes p

divide both binomial coefficients. Hence a counterexample requires a prime-free terminal interval $(n - i, n]$. Further, every integer in $n - i + 1, \dots, n$ must be j -smooth, since a prime $q > j$ dividing one of these integers would again divide both coefficients.

5. LIMITATIONS AND REPRODUCIBILITY

Theorem 1.1 says nothing about $n > 10^8$ at $i = 3$, and nothing about the unresolved general strata $i \geq 4$. A finite negative search cannot justify extrapolation to either region. The accepted partial ranges cited in the introduction are prior work and are not re-claimed here.

The repository contains source, exact output, commands, environment details, malformed-input tests, independent role audits, and a release manifest. The core commands are

```
c++ -O3 -std=c++17 -Wall -Wextra -pedantic \
  code/scan_i3.cpp -o code/scan_i3
c++ -O3 -std=c++17 -Wall -Wextra -pedantic \
  code/verify_i3.cpp -o code/verify_i3
code/scan_i3 8 100000000 > certificates/i3_scan_100m.json
code/verify_i3 certificates/i3_scan_100m.json
python3 code/brute_oracle.py 8 2000
python3 code/verify_release_binding.py --root . \
  --binding certificates/i3_release_binding.json
python3 code/test_release_binding.py
```

The reported run used Apple clang 21.0.0 on arm64 macOS with exact unsigned integer arithmetic. No random seed is present because no randomized step is used.

Computational and AI disclosure. The mathematical reductions, C++ implementations, Python oracle, literature search, audits, and manuscript were produced with OpenAI Codex assistance and then cross-checked by independently derived evaluators as described above. No Lean, Coq, Isabelle, or other proof assistant was used to establish the result. The Lean repository cited during the literature audit was inspected only to distinguish a formalized statement from a proof of the open conjecture.

REFERENCES

1. Thomas F. Bloom, *Erdős problem #699*, Erdős Problems, 2026, Accessed 29 August 2026.
2. E. F. Ecklund, R. B. Eggleton, P. Erdős, and J. L. Selfridge, *On the prime factorization of binomial coefficients*, Journal of the Australian Mathematical Society **26** (1978), no. 3, 257–269.
3. Paul Erdős and George Szekeres, *Some number theoretic problems on binomial coefficients*, Gazette of the Australian Mathematical Society **5** (1978), no. 3, 97–99, MR 80e:10010; Zbl 0401.10003.
4. Cong Lu, *Erdős problem 699 rust scanner*, GitHub repository, 2026, Commit 7d030640beeb6bd; accessed 29 August 2026.

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